

我已經仔細閱讀了你們上傳的「日本簡報_牡蠣殼 的複本-1.pdf」以及「牡蠣殼_EN_Poster.pdf」。這是一個充滿創意且極具環保價值的「循環經濟」專題！你們完美詮釋了「以廢治廢（Using one type of waste to solve another）」的核心精神。

最讓我激賞的是你們在**「實驗結果與討論」中的誠實與科學探究精神。當發現吸光度（**Absorbance**）不降反升時，你們沒有掩蓋數據，反而深入探討了「基質效應（**Matrix effects**）」與果膠色素的干擾，這在國際科展中是展現「極高科學素養與批判性思考」**的絕佳亮點！評審會非常喜歡你們這種實事求是的態度。

為你們量身打造的 7 分鐘全英文演講逐字稿如下。講稿大約 900 字左右，以中等語速（約每分鐘 130 字）報告非常剛好。我加入了雙人輪替（**Tag Team**）的設計與肢體動作建議，幫助你們展現絕佳的團隊默契：

7-Minute English Presentation Transcript

[Slide 1: Title Slide]

(Both speakers stand confidently, smile, and make eye contact with the judges)

Speaker A (Hua-Chun): "Good morning, honorable judges. I am Hua-Chun Hsiao."

Speaker B (Yi-Chieh): "And I am Yi-Chieh Wu. We represent Taipei Municipal Lishan High School."

Speaker A: "Today, we are thrilled to present our research: 'Alchemy of Waste Shells and Fruit Peels: Enhancing Pectin Flocculation and Heavy Metal Removal through Calcium Ion Crosslinking Effects'."

[Slide 2-6: Motivation - The Problem]

(Gesture to the slides showing polluted water and shells)

Speaker A: "Our journey began with two major environmental problems in Taiwan. First, the massive accumulation of discarded oyster shells, which are often left unattended and cause severe odor issues. Second, the increasing turbidity and heavy metal pollution in our lakes and ponds."

Currently, water purification heavily relies on synthetic chemical agents. We asked ourselves a simple but powerful question: **Can we use one type of natural waste to solve another?** Can we replace these chemical agents with low-cost, natural materials?"

[Slide 7-8: Experimental Principles - The Chemistry]

(Show your "brainpower" and understanding of chemical mechanisms)

Speaker B: "To turn waste into a resource, we had to understand the chemistry. Oyster shells are primarily Calcium Carbonate (CaCO_3). Through calcination—heating them to high temperatures—we convert them into Calcium Oxide (CaO) and Calcium Hydroxide (Ca(OH)_2).

This calcined oyster shell powder removes heavy metals, like Copper ions (Cu^{2+}), through three main mechanisms: Chemical Precipitation, Ion Exchange, and Surface Complexation. For instance, the hydroxide ions form Copper Hydroxide precipitates, effectively pulling the toxic metal out of the water."

[Slide 9-10 & Poster: The Pectin Crosslinking Mechanism]

Speaker A: "But removing heavy metals is only half the battle; we also need to clear the turbidity. This is where our second waste material comes in: **Fruit Peels**, specifically their natural Pectin."

(Point to the crosslinking diagram) Here is the 'Alchemy' of our project. Pectin has negative functional groups. When introduced to the Calcium ions (Ca^{2+}) released from our treated oyster shells, a powerful 'Crosslinking Effect' occurs. The Calcium acts as a bridge, pulling the pectin molecules together to form a massive biological net. This 'flocculation' process traps suspended particles, significantly reducing water turbidity."

[Poster: Method & Material]

Speaker B: "To test this, we built a rigorous experimental setup. We dried and ground different fruit peels—specifically Orange, Apple, and Banana. Then, we packed glass columns with different ratios of our oyster shell powder and fruit pectin to perform continuous filtration and adsorption experiments using a 0.05M Copper Sulfate solution."

[Slide 13-14: Experimental Results & The Unexpected Twist]

(Speak with an analytical tone, turning a challenge into a scientific discovery)

Speaker A: "Now, let's look at our results. To measure the removal of Copper ions, we used a spectrophotometer to test the absorbance of the solution. According to the Beer-Lambert law, lower absorbance should mean less Copper. However, we encountered a highly unexpected result. As you can see in our data for Orange, Apple, and Banana pectin, the absorbance *actually increased* from 0.009 to over 0.070 after filtration! Did our filter magically create more heavy metals? Of course not."

Speaker B: "This brings us to a crucial scientific discussion: **Matrix Effects**."

We realized that as the solution passed through the pectin, natural plant pigments and organic matter dissolved into the water. These pigments severely interfered with the spectrophotometer readings at the specific wavelength we were using. Because of this matrix effect, the Beer-Lambert law could not be used directly to back-calculate the decrease in copper ion concentration."

[Poster: Conclusion & Future Applications]

(Transition to the big picture - look directly at the judges)

Speaker A: "Despite this measurement limitation, visual and physical evidence strongly confirms that the composite material formed from oyster shells and pectin has exceptional potential to reduce water turbidity and improve water clarity through calcium crosslinking. Our vision is to manufacture this composite into convenient, capsule-shaped bio-adsorbents. This provides a low-cost, low-environmental-impact solution for water treatment."

Speaker B: "Ultimately, our research perfectly aligns with multiple UN Sustainable Development Goals, including **SDG 6 (Clean Water)**, **SDG 12 (Responsible Consumption and Production)**, and **SDG 15 (Life on Land)**. By combining the calcium from oyster shells with the pectin from fruit peels, we have proven that nature's waste holds the key to healing nature itself."

Thank you very much for your time and attention. We are now happy to answer your questions."

💡 輔導老師的實戰小叮嚀

1. 大方承認測量限制 (**Matrix Effects**)：在 Slide 13-14 講述吸光度上升時，語氣要帶有「科學探究的驚喜感」。評審非常喜歡聽到學生說：「我們一開始預期 A，但卻得到了 B。我們沒有竄改數據，而是去找出為什麼會發生 B。」這會讓你們的專題從單純的「實驗報告」昇華為真正的「科學研究」。
2. 視覺輔助 (**Crosslinking Mechanism**)：當講到鈣離子與果膠的「交聯效應 (Crosslinking Effect)」時，如果可以，請用兩隻手做出「網子 (Net)」或「橋接 (Bridge)」的動作，這能幫助非化學背景的評審瞬間理解你們的核心原理。
3. 專有名詞發音：報告前請多加練習幾個關鍵字彙的咬字與重音，例如：Calcination (煅燒), Flocculation (絮凝), Spectrophotometer (分光光度計), Matrix Effects (基質效應)。

🔍 模擬評審提問 (Judge's Q&A) 預告

由於這個題目牽涉到化學機制與實驗誤差分析，國際科展的評審極可能會針對**「重金屬的替代檢測方案」與「真實環境的應用限制」**提出挑戰。例如：

1. **【關於檢測方法的反思】**"Since the Beer-Lambert law and spectrophotometer failed due to the pigments (matrix effects), how would you accurately measure the remaining Copper ion concentration if you had more time and resources?" (既然分光光度計受色素干擾，如果有更多時間資源，你們會用什麼儀器來精確測量剩下的銅離子？ 提示：可以朝 AAS 原子吸收光譜儀 或 ICP-OES 思考)
2. **【關於果膠選擇】**"You tested orange, apple, and banana peels. Which one had the best flocculation visually, and why do you think different fruits have different effects?"

🎤 國際科展模擬 Q&A：牡蠣殼與果皮的鍊金術 (10 題)

1. 【核心化學機制 The Crosslinking Mechanism】(必考題！)

Q1: You mentioned a "Calcium Ion Crosslinking Effect." Can you explain the exact chemical mechanism of how the calcium from oyster shells interacts with the pectin?

(你們提到了「鈣離子交聯效應」。能解釋牡蠣殼中的鈣是如何與果膠相互作用的確切化學機制嗎？)

- **Suggested Answer:** Certainly. Pectin molecules contain carboxyl groups (-COOH). In a water solution, these groups can become negatively charged (-COO^-). The calcined oyster shells release positively charged Calcium ions (Ca^{2+}). Because Calcium has a +2 charge, a single Calcium ion can bind to two different negatively charged pectin chains simultaneously. This creates a "bridge" or a 3D structural net, pulling the suspended particles together into large flocs that settle easily. This is the core of our flocculation mechanism.
- **中文參考：**沒問題。果膠分子含有羧基 (-COOH)。在水溶液中，這些基團可以帶負電 (-COO^-)。煅燒過的牡蠣殼會釋放帶正電的鈣離子 (Ca^{2+})。因為鈣離子帶 +2 價電荷，單一個鈣離子可以同時與兩條不同的帶負電果膠鏈結合。這產生了一種「橋接」或 3D 結構網，將懸浮顆粒拉在一起形成容易沉澱的大絮凝物。這就是我們絮凝機制的關鍵。
- **🎤 輔導老師提點：**講出 "Carboxyl groups" (羧基) 和 "Bridge" (橋接)，並搭配雙手相扣的手勢，能讓評審立刻明白你們具備扎實的化學底子。

2. 【材料處理 Calcination Process】

Q2: Why did you need to calcine (heat) the oyster shells? Why couldn't you just use raw crushed oyster shells for water purification?

(你們為什麼需要煅燒(加熱)牡蠣殼？為什麼不能直接用壓碎的生牡蠣殼來淨水？)

- **Suggested Answer:** Raw oyster shells are mostly Calcium Carbonate (CaCO_3), which is highly stable and insoluble in water, meaning it releases very few Calcium ions. By calcining them at high temperatures, we convert the CaCO_3 into Calcium Oxide (CaO), which then forms Calcium Hydroxide (Ca(OH)_2) in water. This calcined form is highly reactive, releases abundant Calcium ions for the crosslinking with pectin, and provides the necessary hydroxide (OH^-) ions to precipitate the heavy metals.
- **中文參考：**生牡蠣殼主要是碳酸鈣 (CaCO_3)，它非常穩定且不溶於水，這意味著它釋放的鈣離子極少。透過高溫煅燒，我們將 CaCO_3 轉化為氧化鈣

(CaO)，然後在水中形成氫氧化鈣 (Ca(OH)_2)。這種煅燒後的形式具有高反應性，能釋放大量鈣離子與果膠交聯，並提供必要的氫氧根離子 (OH^-) 來沉澱重金屬。

3. 【實驗誤差與基質效應 Matrix Effects】(大魔王題！)

Q3: Your absorbance data increased after filtration because of the plant pigments. How did you scientifically prove that this was a "matrix effect" and not just your filter generating more heavy metals?

(由於植物色素，你們過濾後的吸光度數據上升了。你們如何科學地證明這是「基質效應」，而不是你們的過濾器產生了更多的重金屬？)

- **Suggested Answer:** We proved this through control experiments and visual evidence. We tested filtering pure distilled water (with zero copper) through our pectin-shell columns, and the water still turned slightly yellow-brown, increasing the absorbance at that specific wavelength. This confirmed that the spectrophotometer was reading the dissolved organic pigments from the fruit peels, not newly generated copper ions. The Beer-Lambert law assumes a single absorbing solute, so the mixed pigments caused a critical matrix interference.
- **中文參考：** 我們透過對照實驗和視覺證據證明了這一點。我們測試將純蒸餾水（零銅離子）過濾通過我們的果膠-牡蠣殼管柱，水依然變成了微黃褐色，增加了該特定波長的吸光度。這證實了分光光度計讀取到的是從果皮溶解出來的有機色素，而不是新產生的銅離子。啤酒-蘭伯特定律（Beer-Lambert law）假設只有單一吸收溶質，因此混合色素造成了嚴重的基質干擾。

4. 【替代檢測方案 Alternative Measurement】(必準備！)

Q4: Since the spectrophotometer failed due to the pigments, what instrument should you use to accurately measure the remaining Copper ion concentration if you continue this research?

(既然分光光度計因為色素而失效，如果繼續這項研究，你們「應該」使用什麼儀器來準確測量剩餘的銅離子濃度？)

- **Suggested Answer:** That is our immediate next step. To overcome the color and organic matrix interference, we need to use an Atomic Absorption Spectrophotometer (AAS) or Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES). These instruments atomize the sample at extremely high temperatures and specifically measure the concentration of the Copper element itself, completely ignoring any organic pigments or color in the water.
- **中文參考：** 這是我們眼下的下一步。為了克服顏色和有機基質的干擾，我們需要使用原子吸收光譜儀 (AAS) 或 感應耦合電漿光學發射光譜儀 (ICP-OES)。這些儀器會在極高溫下將樣品原子化，並專門測量銅元素本身的濃度，完全忽略水中的任何有機色素或顏色。
- **💡 輔導老師提點：** 講出 **AAS** 或 **ICP-OES**，評審會對你們的儀器分析知識刮目相看！這證明你們不僅知道錯在哪，還知道怎麼解決。

5. 【變因比較 Fruit Peel Comparison】

Q5: You tested Orange, Apple, and Banana peels. Based on your visual observations of flocculation, which fruit peel worked the best and why do you think that is?

(你們測試了橘子、蘋果和香蕉皮。根據你們對絮凝的視覺觀察，哪種果皮效果最好？你們認為原因是什麼？)

- **Suggested Answer:** Based on our visual observation, the Orange peel powder generally exhibited the most effective and rapid flocculation. We believe this is because citrus peels (like orange) naturally contain a higher concentration of pectin and possess different degrees of esterification compared to apples or bananas. The

denser carboxyl groups in orange pectin provide more active binding sites for the Calcium ions to form a stronger crosslinked network.

- **中文參考：** 根據我們的視覺觀察，橘子皮粉末通常表現出最有效且快速的絮凝效果。我們認為這是因為與蘋果或香蕉相比，柑橘類果皮（如橘子）天然含有更高濃度的果膠，且具有不同的酯化程度。橘子果膠中更密集的羧基為鈣離子提供了更多活躍的結合位點，從而形成更強的交聯網絡。

6. 【潛在的二次污染 Secondary Pollution】

Q6: Using calcined oyster shells (\$Ca(OH)₂) will raise the pH of the water, making it highly alkaline. Furthermore, fruit peels introduce organic matter. Won't your solution cause secondary pollution to the lake?

(使用煅燒牡蠣殼會提高水的 pH 值，使其呈強鹼性。此外，果皮會引入有機物。你們的解決方案不會對湖泊造成二次污染嗎？)

- **Suggested Answer:** That is a very critical engineering consideration. You are right; directly dumping this raw powder into a lake could cause an alkaline shock and increase the Biological Oxygen Demand (BOD). That is why our future application is not loose powder, but a **"Capsule-shaped Bio-adsorbent."** By pre-crosslinking the materials and enclosing them in a permeable, biodegradable hydrogel capsule, we can control the release rate of the ions and easily retrieve the capsules after the heavy metals are adsorbed, preventing secondary pollution.
- **中文參考：** 這是一個非常關鍵的工程考量。您說得對；直接將這些生粉末倒入湖中可能會造成鹼性衝擊並增加生化需氧量 (BOD)。這就是為什麼我們未來的應用不是散裝粉末，而是**「膠囊狀生物吸附劑」**。透過預先交聯材料並將其封裝在具滲透性、可生物降解的水凝膠膠囊中，我們可以控制離子的釋放速率，並在重金屬被吸附後輕鬆回收膠囊，防止二次污染。

7. 【工程權衡 Trade-offs & Energy Cost】

Q7: You emphasize that this is a "low-cost, eco-friendly" solution. However, calcining oyster shells requires heating them to very high temperatures (often over 800°C). Doesn't the massive energy cost and carbon emission contradict your eco-friendly goal?

(你們強調這是一個「低成本、環保」的解決方案。然而，煅燒牡蠣殼需要將其加熱到非常高的溫度。龐大的能源消耗和碳排放難道不違背你們環保的目標嗎？)

- **Suggested Answer:** You have pinpointed a major challenge in the circular economy. Calcination is indeed energy-intensive. However, to maintain our eco-friendly goal on an industrial scale, we propose integrating this process with existing heavy industries. For example, the oyster shells could be calcined using the "waste heat" generated by steel mills or municipal incinerators. By utilizing industrial waste heat, we can transform the shells without consuming additional fossil fuels, maintaining a true zero-carbon circular loop.
- **中文參考：** 您準確指出了循環經濟中的一個重大挑戰。煅燒確實是高耗能的。然而，為了在工業規模上維持我們環保的目標，我們建議將此過程與現有的重工業整合。例如，可以使用煉鋼廠或市政焚化爐產生的「廢熱」來煅燒牡蠣殼。透過利用工業廢熱，我們可以在不消耗額外化石燃料的情況下轉化牡蠣殼，維持真正的零碳循環迴路。

8. 【重金屬移除機制 Heavy Metal Removal】

Q8: Besides chemical precipitation (forming Copper Hydroxide), your poster mentions "Ion Exchange" and "Surface Complexation." Can you briefly explain what these are?

(除了化學沉澱外，你們的海報提到了「離子交換」和「表面錯合」。能簡短解釋一下這是什麼嗎？)

- **Suggested Answer:** Certainly. "Ion Exchange" happens when the Calcium ions (Ca^{2+}) on the surface of the shell matrix are physically swapped with the toxic Copper ions (Cu^{2+}) in the water. "Surface Complexation" occurs when the Copper ions form strong chemical bonds directly with the oxygen atoms present on the surface of the calcined shell structure. Together with precipitation, these three mechanisms create a highly efficient multi-layered heavy metal trap.
- **中文參考：**當然。「離子交換」發生在殼基質表面的鈣離子 (Ca^{2+}) 與水中的有毒銅離子 (Cu^{2+}) 進行物理置換時。「表面錯合」則是當銅離子直接與存在於煅燒殼結構表面的氧原子形成強大的化學鍵時發生。這三種機制與沉澱作用結合，創造了一個高效的多層重金屬陷阱。

9. 【永續影響力 SDG Impact】

Q9: How exactly does this project contribute to the United Nations' SDG 12 (Responsible Consumption and Production)?

(這個專案究竟是如何對聯合國的 SDG 12 (負責任的消費與生產) 做出貢獻?)

- **Suggested Answer:** SDG 12 focuses on "doing more and better with less" and drastically reducing waste. Our project is the perfect embodiment of this. We take two major agricultural waste streams in Taiwan—discarded oyster shells and fruit peels—and upcycle them into a valuable water treatment product. This prevents waste from entering landfills and replaces the need to manufacture new, synthetic chemical flocculants, fully realizing the principle of a circular economy.
- **中文參考：**SDG 12 的重點是「用更少的資源做更多更好的事」，並大幅減少廢棄物。我們的專案正是此理念的完美體現。我們採用了台灣兩大農業廢棄物（廢棄牡蠣殼和果皮），將其升級回收為有價值的淨水產品。這防止了廢棄物進入垃圾掩埋場，並取代了製造新合成化學絮凝劑的需求，完全實現了循環經濟的原則。

10. 【研究挑戰 The Biggest Challenge】

Q10: What was the most frustrating moment during this research, and what did you learn from it?

(在這個研究過程中，最令你們挫折的時刻是什麼？你們從中學到了什麼？)

- **Suggested Answer:** The most frustrating moment was definitively when we looked at the spectrophotometer data and saw the absorbance *increasing* after filtration. At first, we thought our entire experiment had failed. But we learned that in science, anomalous data is often a disguised opportunity. Instead of ignoring it, we dug into analytical chemistry and discovered the "matrix effect" of plant pigments. We learned that a true scientist must question the measuring tool itself, not just blindly trust the numbers.
- **中文參考：**最令人挫折的時刻絕對是當我們看著分光光度計的數據，發現過濾後吸光度竟然「上升」的時候。起初，我們以為整個實驗都失敗了。但我們學到，在科學中，異常數據通常是偽裝的機會。我們沒有忽略它，而是深入研究分析化學，發現了植物色素的「基質效應」。我們學到，一個真正的科學家必須質疑測量工具本身，而不是盲目相信數字。

💡 實戰排練建議

這 10 題包含了非常多化學工程的經典問題。特別是 **Q3 (基質效應)**、**Q4 (AAS 儀器)** 與 **Q6 (二次污染)**，請 Hua-Chun 和 Yi-Chieh 務必熟練這幾題的答辯邏輯。

如果遇到外籍評審問的問題很長、聽不清楚，可以禮貌地微笑說：

"Thank you for the insightful question. Just to ensure I fully understand, are you asking about [重述你聽懂的關鍵字, 例如: the secondary pollution]?"